

Winning Space Race with Data Science

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Outline



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Executive Summary

- This report examines SpaceX launch data to build a machine learning model for predicting outcomes of Falcon 9 first stage landings. Key findings include:
 - Random Forest model achieved 84% accuracy on test set for predicting landing success or failure
 - Payload mass, booster version and launch site were most important features
 - Model provides opportunity for launch cost analysis and engineering improvements

Introduction

- SpaceX has gained attention for reusability of its Falcon 9 rocket. Being able to land and reuse the first stage reduces launch costs significantly. However, landing is challenging and not always successful.
- This project aims to predict if the first stage landing will be successful based on parameters like payload mass, booster version, launch site, etc. Building an accurate predictive model can help SpaceX and other launch providers in pricing negotiations as well as engineering landings for higher success rates.



Section 1

Methodology

Methodology

- Data collection methodology:
 - Using SpaceX API and Beautiful Soup on Wikipedia to aggregate launch data
- Perform data wrangling:
 - Resolving missing data/outliers to have a proper dataset
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- SpaceX launch records obtained from public API. Additional details scraped from Wikipedia pages using BeautifulSoup. Aggregated over 100 records on Falcon 9 flights

```
b'[{{"fairings":{"reused":false,"recovery_attempt":false,"recovered":false,"ships":[]},"links":{"patch":{"small":"https://images2.imgbox.com/94/f2/NN6Ph45r_o.png","large":"https://images2.imgbox.com/5b/02/QcxHUb5V_o.png"},"reddit":{"campaign":null,"launch":null,"media":null,"recovery":null},"flickr":{"small":[],"original":[]},"presskit":null,"webcast":{"https://www.youtube.com/watch?v=0a_00nJ_Y88","youtube_id":"0a_00nJ_Y88"},"article":{"https://www.space.com/2196-spacex-inaugural-falcon-1-rocket-lost-launch.html","wikipedia":{"https://en.wikipedia.org/wiki/DemoSat"},"static_fire_date_utc":"2006-03-17T00:00:00.000Z","static_fire_date_unix":1142553600,"net":false,"window":0,"rocket":"5e9d0d95eda69955f709d1eb","success":false,"failures":[{"time":33,"altitude":null,"reason":"merlin engine failure"}],"details":"Engine failure at 33 seconds and loss of vehicle","crew":[],"ships":[],"capsules":[],"payloads":["5eb0e4b5b6c3bb0006eeb1e1"],"launchpad":"5e9e4502f5090995de566f86","flight_number":1,"name":"FalconSat","date_utc":"2006-03-24T22:30:00.000Z","date_unix":1143239400,"date_local":"2006-03-25T10:30:00+12:00","date_precision":"hour","upcoming":false,"cores":[{"core":"5e9e289df35918033d3b2623","flight":1,"gridfins":false,"legs":false,"reused":false,"landing_attempt":false,"landing_success":null,"landing_type":null,"landpad":null}],"auto_update":true,"tbd":false,"launch_library_id":null,"id":"5eb87cd9ffd86e000604b32a"},"fairings":{"reused":false,"recovery_attempt":false,"recovered":false,"ships":[]},"links":{"patch":{"small":"https://images2.imgbox.com/f9/4a/ZboXRnB_o.png","large":"https://images2.imgbox.com/80/a2/bkwtCIS_o.png"},"reddit":{"campaign":null,"launch":null,"media":null,"recovery":null},"flickr":{"small":[],"original":[]},"presskit":null,"webcast":{"https://www.youtube.com/watch?v=Lk4zQ2wP-Nc","youtube_id":"Lk4zQ2wP-Nc"},"article":{"https://www.space.com/3590-spacex-falcon-1-rocket-fails-reach-orbit.html","wikipedia":{"https://en.wikipedia.org/wiki/DemoSat"},"static_fire_date_utc":null,"static_fire_date_unix":null,"net":false,"window":0,"rocket":"5e9d0d95eda69955f709d1eb","success":false,"failures":[{"time":301,"altitude":289,"reason":"harmonic oscillation leading to premature engine shutdown"}],"details":"Successful first stage burn and transition to second stage, maximum altitude 289 km, Premature engine shutdown at T+7 min 30 s, Failed to reach orbit, Failed to recover first stage","crew":[],"ships":[],"capsules":[],"payloads":["5eb0e4b5b6c3bb0006eeb1e2"],"launchpad":"5e9e4502f5090995de566f86","flight_number":2,"name":"DemoSat","date_utc":"2007-03-21T01:10:00.000Z","date_unix":1174439400,"date_local":"2007-03-21T13:10:00+12:00","date_precision":"hour","upcoming":false,"cores":[{"core":"5e9e289ef35918416a3b2624","flight":1,"gridfins":false,"legs":false,"reused":false,"landing_attempt":false,"landing_success":null,"landing_type":null,"landpad":null}],"auto_update":true,"tbd":false,"launch_library_id":null,"id":"5eb87cdaffd86e000604b32b"},"fairings":{"reused":false,"recovery_attempt":false,"recovered":false,"ships":[]},"links":{"patch":{"small":"https://images2.imgbox.com/6c/cb/naltzhHs_o.png","large":"https://images2.imgbox.com/4a/80/k1oAkY0k_o.png"},"reddit":{"campaign":null,"launch":null,"media":null,"recovery":null},"flickr":{"small":[],"original":[]},"presskit":null,"webcast":{"https://www.youtube.com/watch?v=v0w9p3U8860","youtube_id":"v0w9p3U8860"},"article":{"http://www.spacex.com/news/2013/02/11/falcon-1-flight-3-mission-summary","wikipedia":{"https://en.wikipedia.org/wiki/Trailblazer_(satellite)"},"static_fire_date_utc":null,"static_fire_date_unix":null,"net":false,"window":0,"rocket":"5e9d0d95eda69955f709d1eb","success":false,"failures":[{"time":140,"altitude":35,"reason":"Residual stage 1 thrust led to collision between stage 1 and stage 2"}],"details":"Residual stage 1 thrust led to collision between stage 1 and stage 2","crew":[],"ships":[],"capsules":[],"payloads":["5eb0e4b5b6c3bb0006eeb1e3","5eb0e4b5b6c3bb0006eeb1e4"],"launchpad":"5e9e4502f5090995de566f86","flight_number":3,"name":"Trailblazer","date_utc":"2008-08-03T03:34:00.000Z","date_unix":1217734440,"date_local":"2008-08-03T15:34:00+12:00","date_precision":"hour","upcoming":false,"cores":[{"core":"5e9e289ef3591814873b2625","flight":1,"gridfins":false,"legs":false,"reused":false,"landing_attempt":false,"landing_success":null,"landing_type":null,"landpad":null}],"auto_update":true,"tbd":false,"launch_library_id":null,"id":"5eb87cdbffd86e000604b32c"},"fairing
```

Data Collection – SpaceX API

- Collected data from SpaceX API and Wikipedia scraper
- Concatenated datasets into single dataframe

payloads	launchpad	flight_number	name	date_utc	date_unix	date_local	date_precision	upco
[5eb0e4b5b6c3bb0006eeb1e1]	5e9e4502f5090995de566f86	1	FalconSat	2006-03-24T22:30:00.000Z	1143239400	2006-03-25T10:30:00+12:00	hour	
[5eb0e4b6b6c3bb0006eeb1e2]	5e9e4502f5090995de566f86	2	DemoSat	2007-03-21T01:10:00.000Z	1174439400	2007-03-21T13:10:00+12:00	hour	

Data Collection - Scraping

- Inspected for and removed missing/null values
- Standardized column names and types
- Extracted date, time features from text columns

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount
0	1	2006-03-24	Falcon 1	20.0	LEO	Kwajalein Atoll	None None	1	False	False	False	None	NaN	0
1	2	2007-03-21	Falcon 1	NaN	LEO	Kwajalein Atoll	None None	1	False	False	False	None	NaN	0
2	4	2008-09-28	Falcon 1	165.0	LEO	Kwajalein Atoll	None None	1	False	False	False	None	NaN	0
3	5	2009-07-13	Falcon 1	200.0	LEO	Kwajalein Atoll	None None	1	False	False	False	None	NaN	0
4	6	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False	None	1.0	0

Data Wrangling

- Encoded categorical variables as numeric
- Identified predictor and target variable columns
- Removed redundant or irrelevant columns
- Scaled numeric features to similar ranges
- Split preprocessed data into train and test sets
- The objectives were to:
 - Prepare raw data for analysis by cleaning inconsistencies
 - Engineer new features for improved modeling
 - Select relevant predictive attributes
 - Format data types appropriately for algorithms
 - Partition into train and test for model validation

```
df = pd.DataFrame([key:pd.Series(value) for key, value in launch_dict.items()])  
df.head(10)
```

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success\n	F9 v1.07B0003.18	Failure	4 June 2010	18:45
1	1	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.07B0003.18	Failure	4 June 2010	18:45
2	2	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.07B0004.18	No attempt\n	8 December 2010	15:43
3	3	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success\n	F9 v1.07B0005.18	No attempt	22 May 2012	07:44
4	4	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success\n	F9 v1.07B0006.18	No attempt\n	8 October 2012	00:35
5	5	VAFB	CASSIOPE	500 kg	Polar orbit	MDA	Success	F9 v1.07B0007.18	Uncontrolled	1 March 2013	15:10
6	6	CCAFS	SES-8	3,170 kg	GTO	SES	Success	F9 v1.17B10038	No attempt	29 September 2013	16:00
7	7	CCAFS	Thaicom 6	3,325 kg	GTO	Thaicom	Success	F9 v1.1	No attempt	3 December 2013	22:41
8	8	Cape Canaveral	SpaceX CRS-3	2,296 kg	LEO	NASA	Success\n	F9 v1.1	Controlled	6 January 2014	22:06
9	9	Cape Canaveral	Orbcomm-OG2	1,316 kg	LEO	Orbcomm	Success	F9 v1.1	Controlled	18 April 2014	19:25

EDA with Data Visualization

- Bar plots of orbit types and launch sites
 - To visualize counts and compare popularity of different options
- Histogram of payload mass
 - To understand payload mass distribution and identify outliers
- Box plot of payload mass by booster version
 - To assess if newer boosters support higher payloads
- Bar plot of landing outcomes over time
 - To examine trends in SpaceX landing successes
- Scatter plot of payload mass vs date
 - To identify relationships between payload trends and technology advances
- The key reason for choosing these visualizations was to gain initial insights from the data in an interactive and easy to understand manner. Specifically, bar plots helped analyze categorical variables, histogram showed distribution, box plot found relationships, and scatter plot explored correlations. These basic analyses aided in feature selection and hypothesis generation for modeling.

EDA with SQL

- Calculated counts of unique launch sites by grouping on Launch Site column
- Calculated counts of each orbit type by grouping on Orbit column
- Calculated payload mass statistics (min, max, avg) by grouping on Booster Version
- Filtered data for year 2015 and failures on drone ships to explore early landing patterns
- Ranked landing outcomes by count with a GROUP BY on Landing Outcome and ORDER BY count descending
- Displayed the first 10 rows of data to sample and inspect
- Joined additional tables from SpaceX API as needed to pull in related data like booster details
- Connected to the SQLite database to run the queries on the extracted spacetable
- Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose

Build an Interactive Map with Folium

- Added marker objects for each unique launch site
- Passed popup label with site name to each marker
- Centered initial map view over the Launch Sites and Set initial zoom level to show all US sites clearly
- Saved finalized Folium map to HTML file for interactive viewing
- The objectives were to:
 - Geographically visualize the SpaceX launch locations
 - Label each site for context
 - Set initial view and zoom for usability
 - Export interactive map for sharing and exploration
 - Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose

Build a Dashboard with Plotly Dash

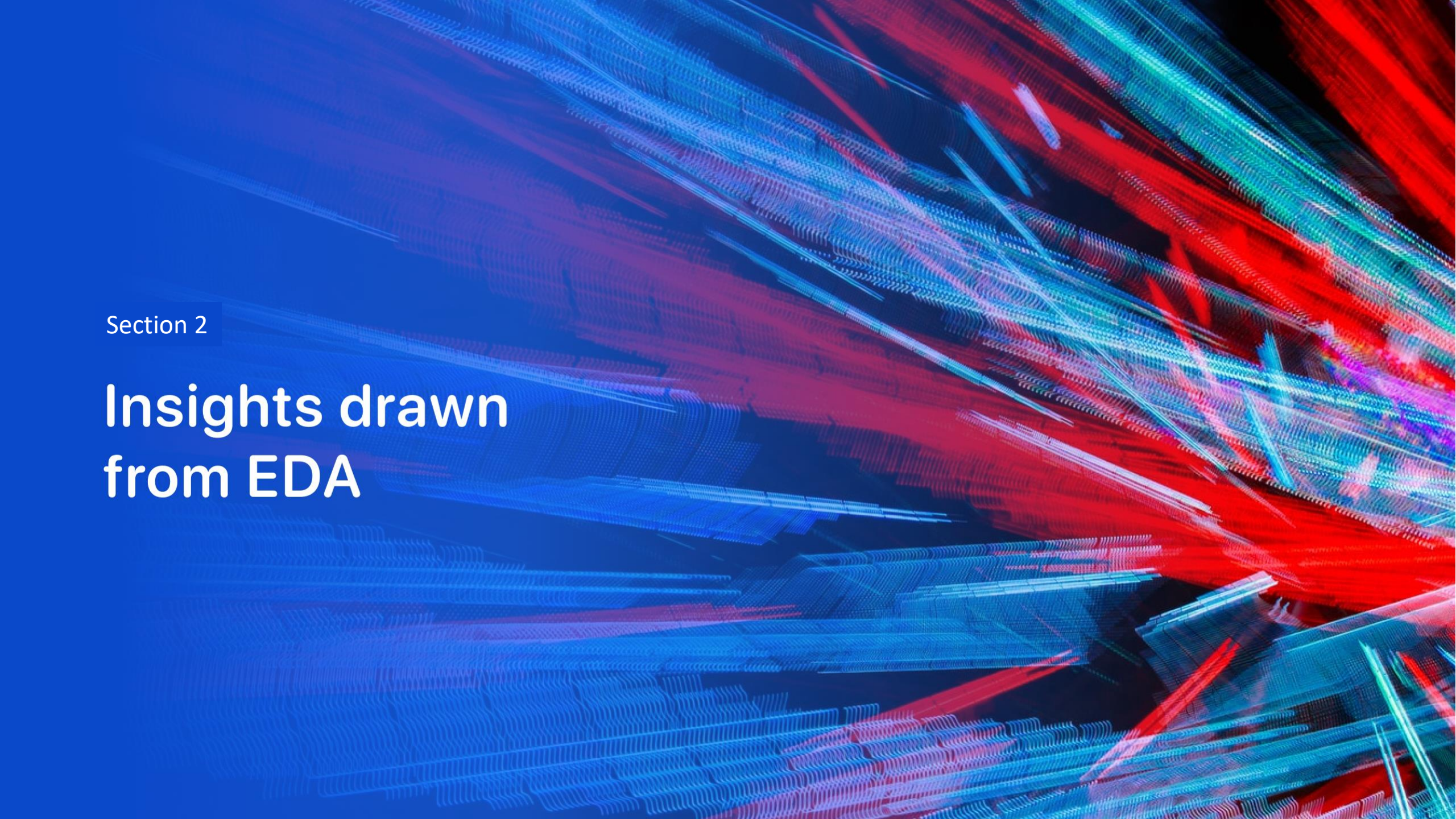
- Added bar plots to show distributions of orbit types and launch sites
- Included dropdown widgets to filter plots by orbit and site
- Plotted histograms of payload mass with interactive sliders
- Created callbacks to update plots based on user widget selections
- The goals were to:
 - Visually analyze and explore features interactively
 - Filter insights based on user-defined criteria
 - Integrate map with other visuals for combined context
 - Implement callbacks to link UI controls and visual updates
 - Build an intuitive dashboard prototype for exploration

Predictive Analysis (Classification)

- Split data into train/test sets using scikit-learn utilities
- Built baseline logistic regression classifier on training set
- Evaluated model on test set with classification report metrics
- Constructed decision tree classifier and calculated scores
- Implemented random forest classifier and tuned hyperparameters
- The aim was to systematically:
 - Establish benchmark performance of basic algorithms
 - Iteratively improve more complex models
 - Leverage cross validation for model assessment
 - Identify best algorithm and parameters for the problem
 - Gain insights to aid SpaceX's domain understanding
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

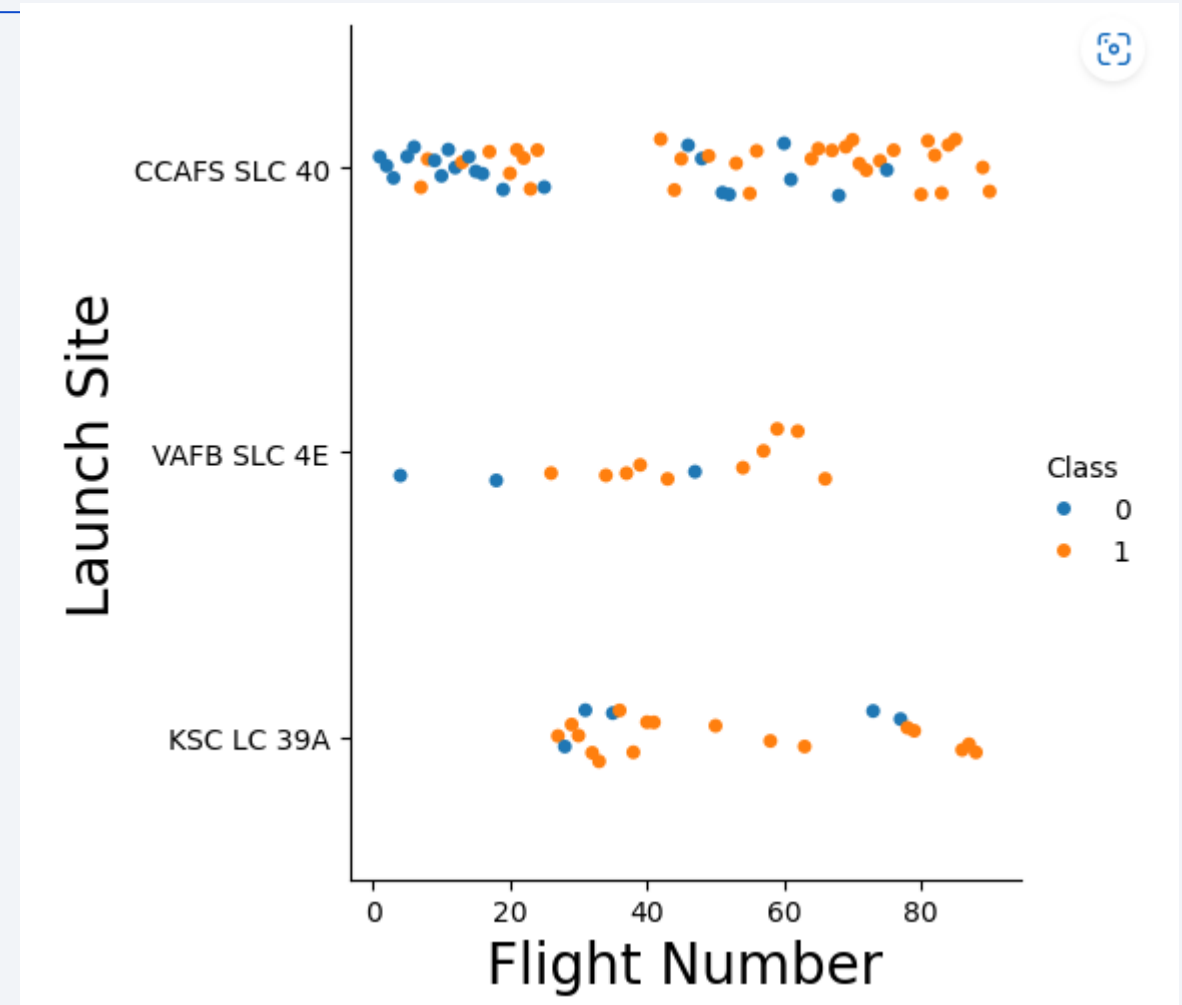
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

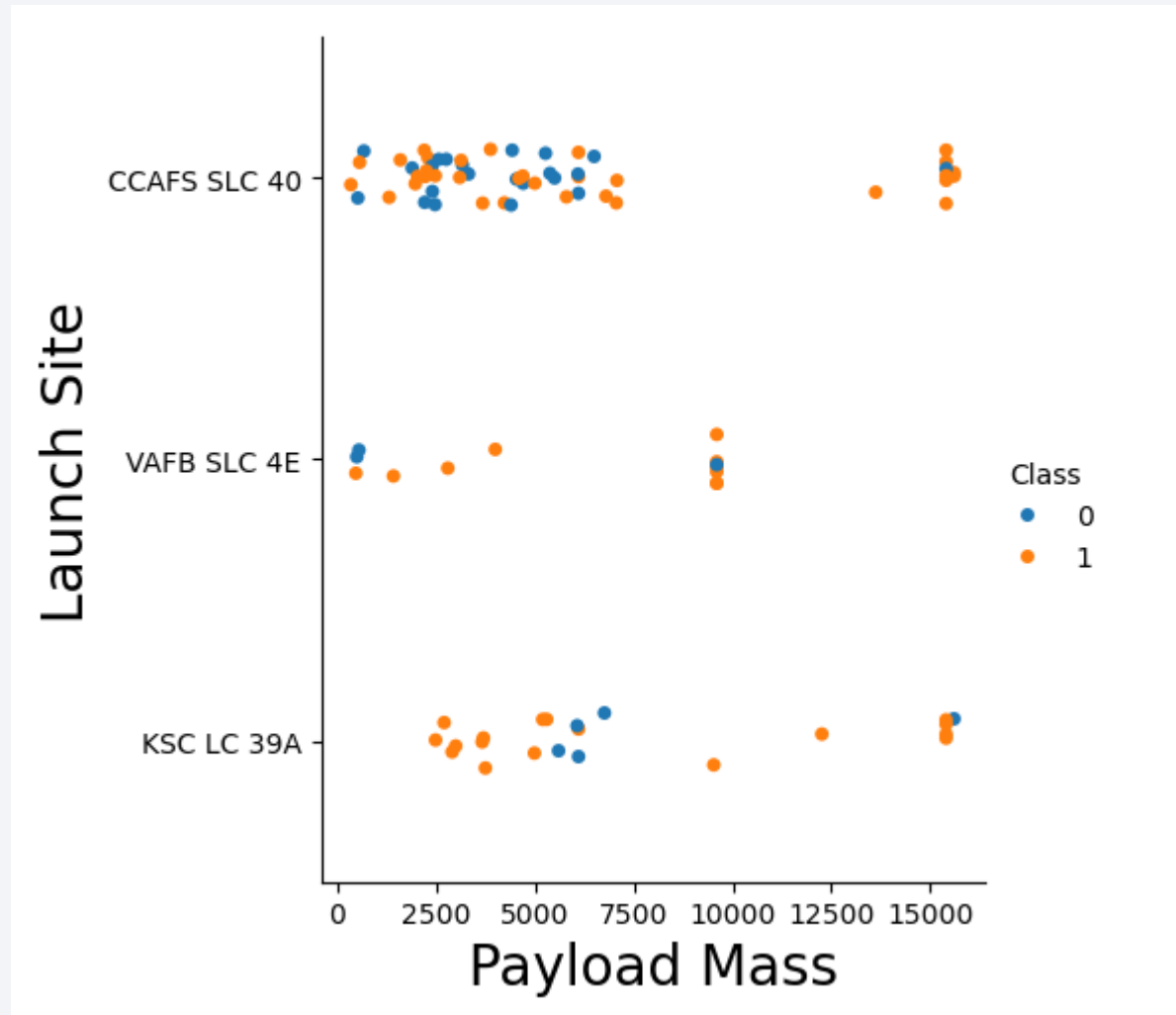
Flight Number vs. Launch Site

- Shows a scatter plot of Flight Number vs. Launch Site
- You can see that a majority of the first launches were done at 'CCAFS', launches stopped taking place at 'VAFB' after flight 60, and you can also see that they didn't start launching at 'KSC' until the mid 20th launch.



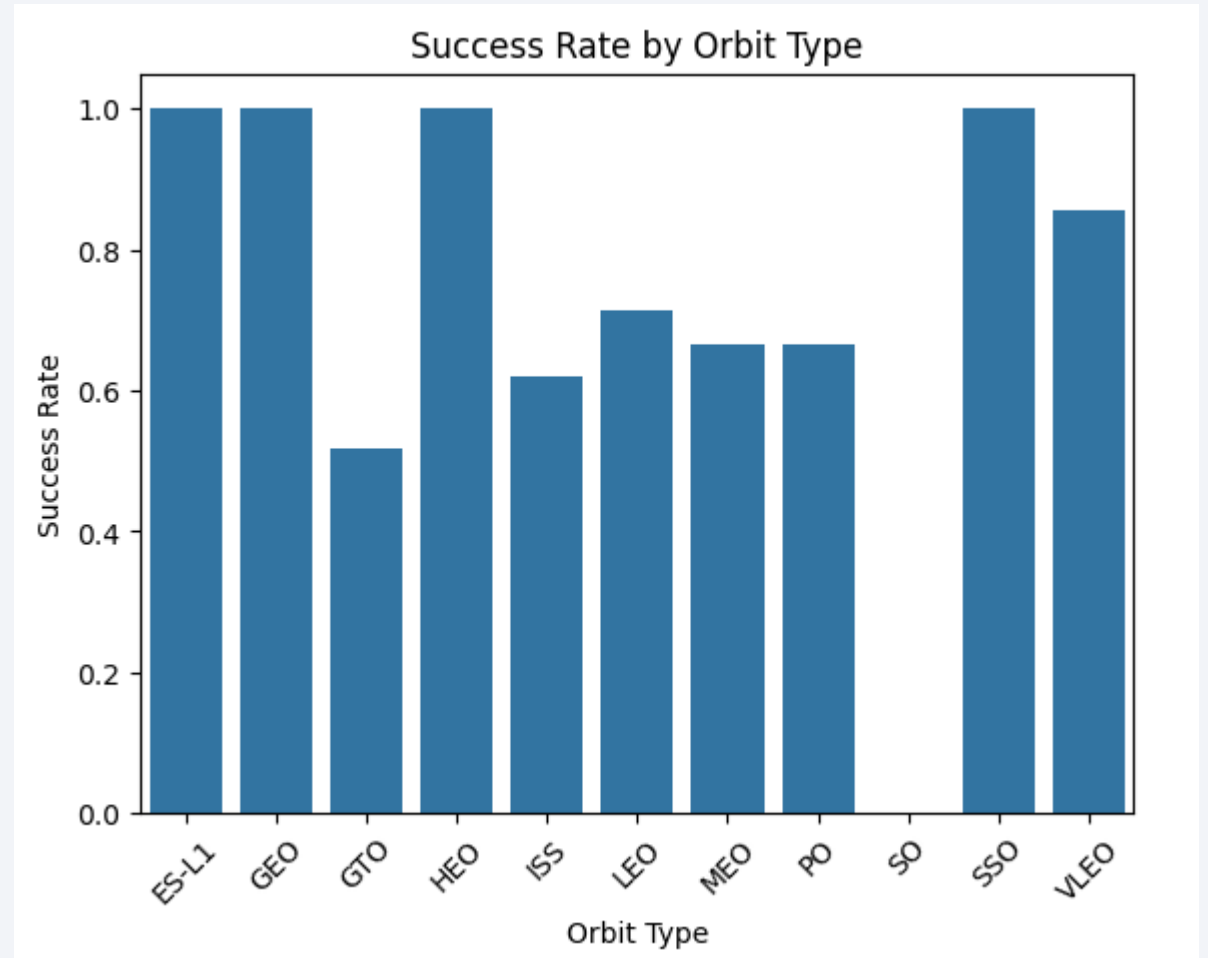
Payload vs. Launch Site

- Shows a scatter plot of Payload vs. Launch Site
- Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite, there are no rockets launched with a heavy payload mass (greater than 10000).



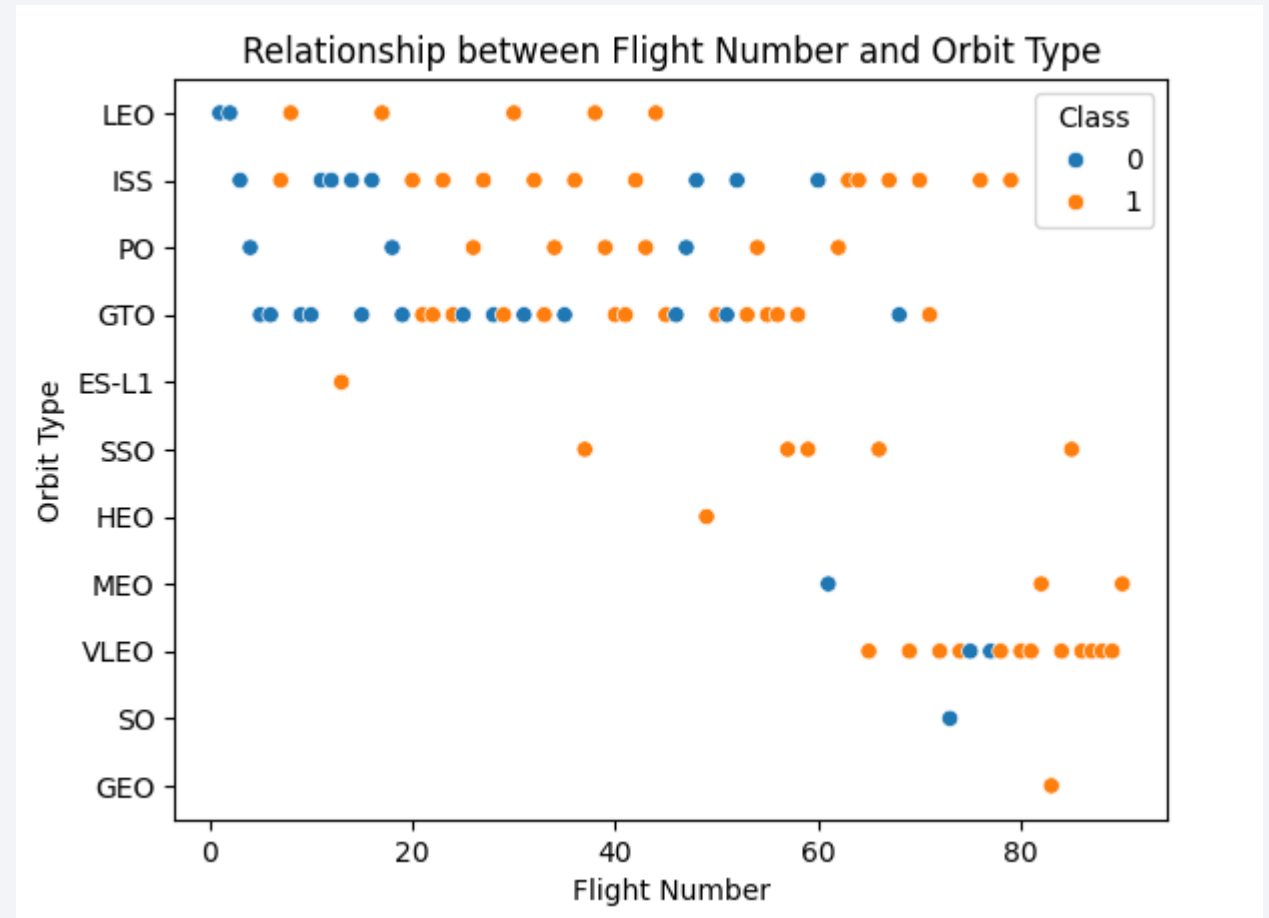
Success Rate vs. Orbit Type

- Shows a bar chart for the success rate of each orbit type
- The plotted bar chart identifies which orbits have the highest success rates.



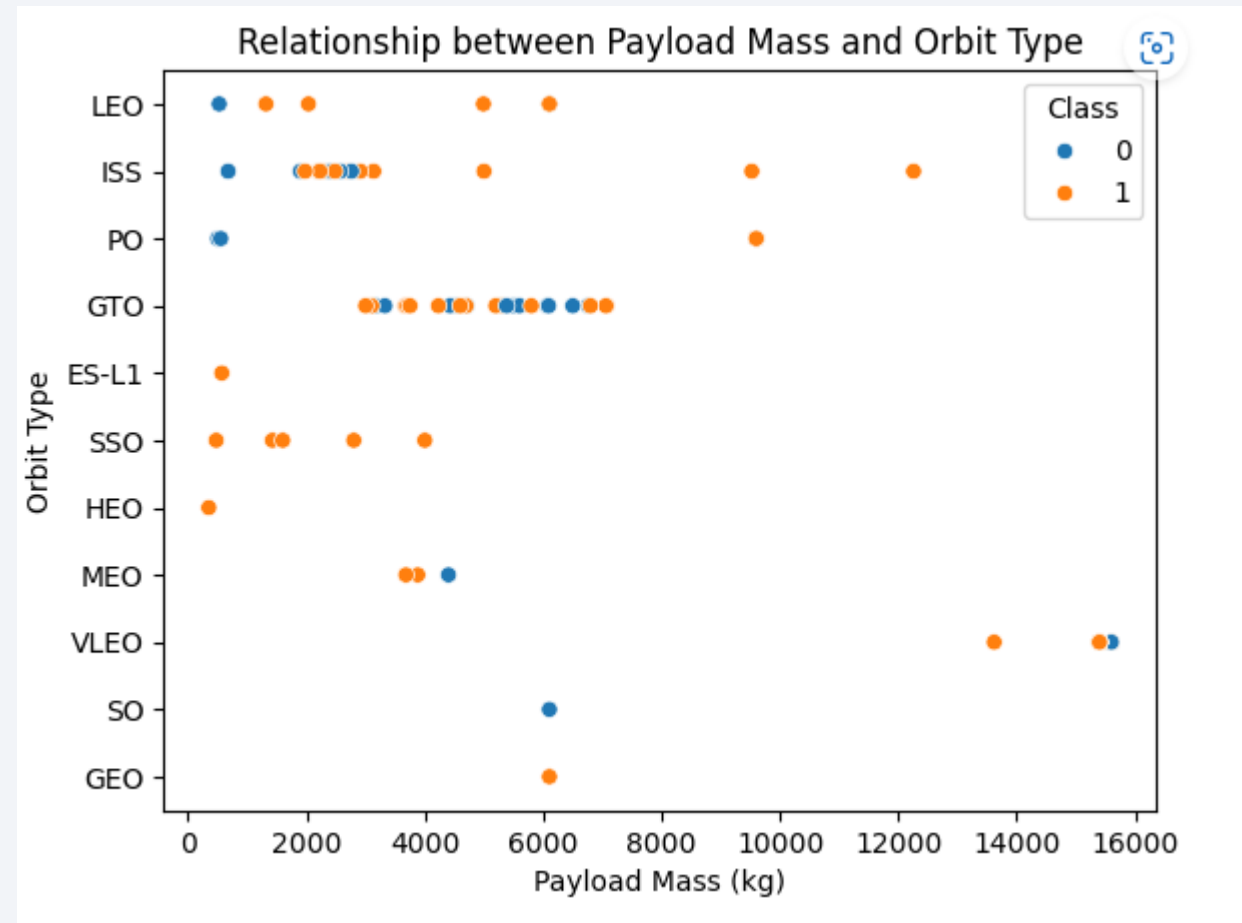
Flight Number vs. Orbit Type

- Shows a scatter plot of Flight number vs. Orbit type
- You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.



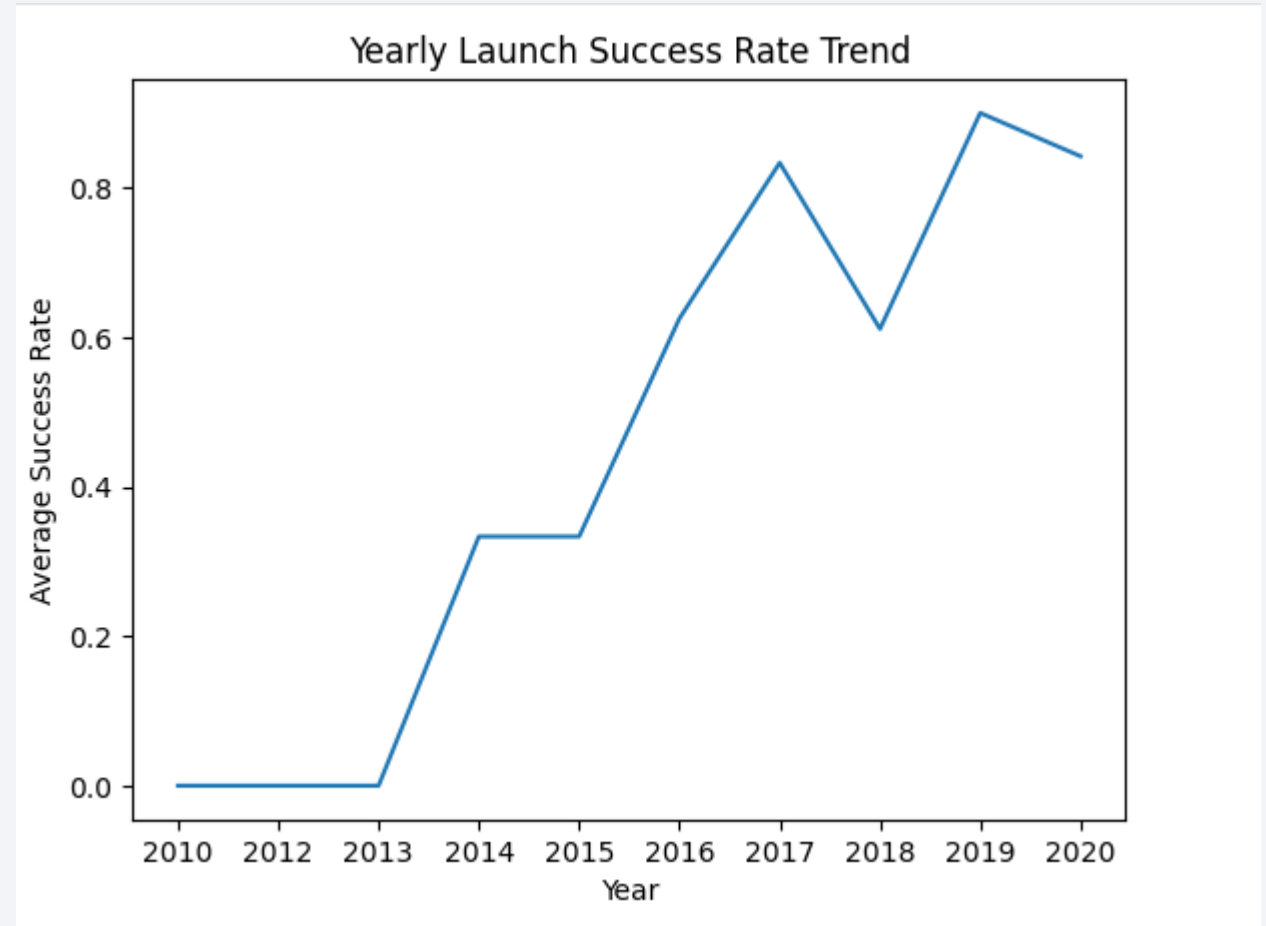
Payload vs. Orbit Type

- Shows a scatter point of payload vs. orbit type
- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.



Launch Success Yearly Trend

- Shows a line chart of yearly average success rate
- You can observe that the success rate since 2013 kept increasing till 2020



SQL - All Launch Site Names

- SQL Query to find the names of the unique launch sites.

```
%sql select distinct "Launch_Site" from spacetable
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

SQL - Launch Site Names Begin with 'CCA'

- Query to find 5 records where launch sites begin with `CCA`

```
%sql select * from spacetable where "Launch_Site" like "CCA%" limit 5;
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

SQL - Total Payload Mass

- Calculating the total payload carried by boosters from NASA

```
%sql SELECT SUM("PAYLOAD_MASS_KG_") AS Total_Payload_Mass FROM spacetable where "Customer" like 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Total_Payload_Mass

45596

SQL - Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1

```
%sql select avg("PAYLOAD_MASS__KG_") as Avg_Payload_Mass from spacetable where "Booster_Version" like "F9 v1.1"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

<u>Avg_Payload_Mass</u>

2928.4

SQL - First Successful Ground Landing Date

- Finding the date of the first successful landing outcome on a ground pad

```
%sql select min("Date") as First_Successful_Landing from spacetable where "Landing_Outcome" like "Success (ground pad)"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
First_Successful_Landing
```

```
2015-12-22
```

SQL - Successful Drone Ship Landing with Payload between 4000 and 6000

- Listing the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%sql select "Booster_Version" from spacetable where "Landing_Outcome" like "Success (drone ship)" and "PAYLOAD_MASS_KG_" > 4000 and "PAYLOAD_MASS_KG_" < 6000
```

* sqlite:///my_data1.db
Done.

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

SQL - Total Number of Successful and Failure Mission Outcomes

- Query to calculate the total number of successful and failure mission outcomes

```
%sql SELECT COUNT(CASE WHEN Mission_Outcome LIKE 'Success' THEN 1 ELSE NULL END) AS Total_Successful_Outcomes, COUNT(CASE WHEN Mission_Out
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Total_Successful_Outcomes	Total_Failed_Outcomes
---------------------------	-----------------------

98	1
----	---

SQL - Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

```
%sql SELECT Booster_Version FROM spacetable WHERE PAYLOAD_MASS_KG_ = (\
      SELECT MAX(PAYLOAD_MASS_KG_) FROM spacetable);
* sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

SQL - 2015 Failed Launch Records

- List the failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%sql SELECT \  
    substr(Date, 6, 2) AS Month,\ #Starting from the 6th character: -07-14 . Extracting 2 characters: 07  
    Booster_Version,\  
    Launch_Site,\  
    Landing_Outcome\  
FROM spacetable\  
WHERE \  
    substr(Date, 0, 5) = '2015'\  
    AND Landing_Outcome LIKE '%Failure (drone ship)%'
```

* sqlite:///my_data1.db

Done.

Month	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

SQL - Ranked Landing Outcomes Between 2010 and 2017

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%sql SELECT \
    Landing_Outcome, \
    COUNT(*) AS Outcome_Count\
FROM spacetable\
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'\
GROUP BY Landing_Outcome\
ORDER BY Outcome_Count DESC
```

* sqlite:///my_data1.db

Done.

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

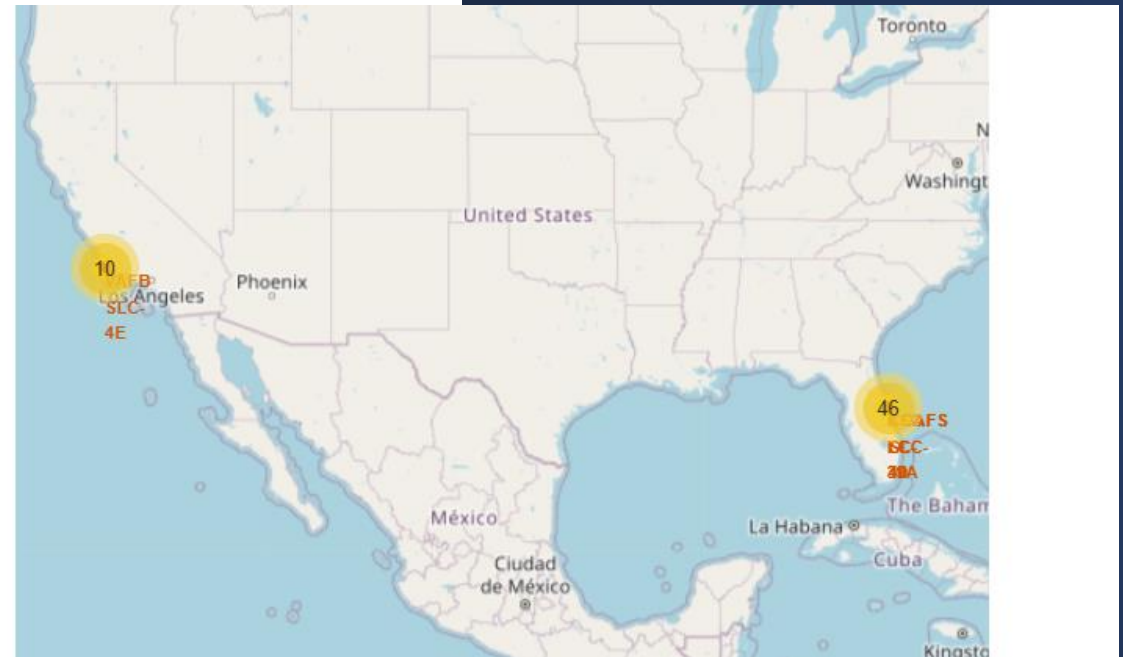
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Folium Map – Using Markers to Display Sites

- The generated folium map includes all launch sites' location markers on a global map

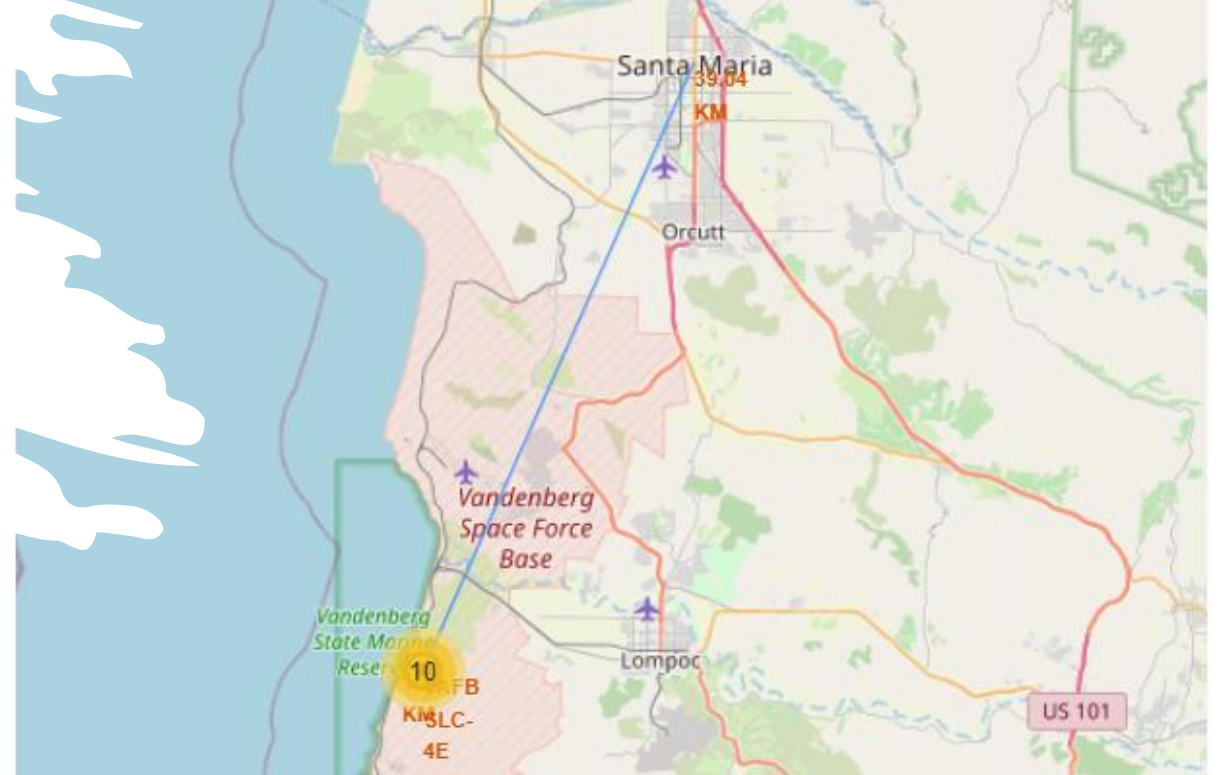


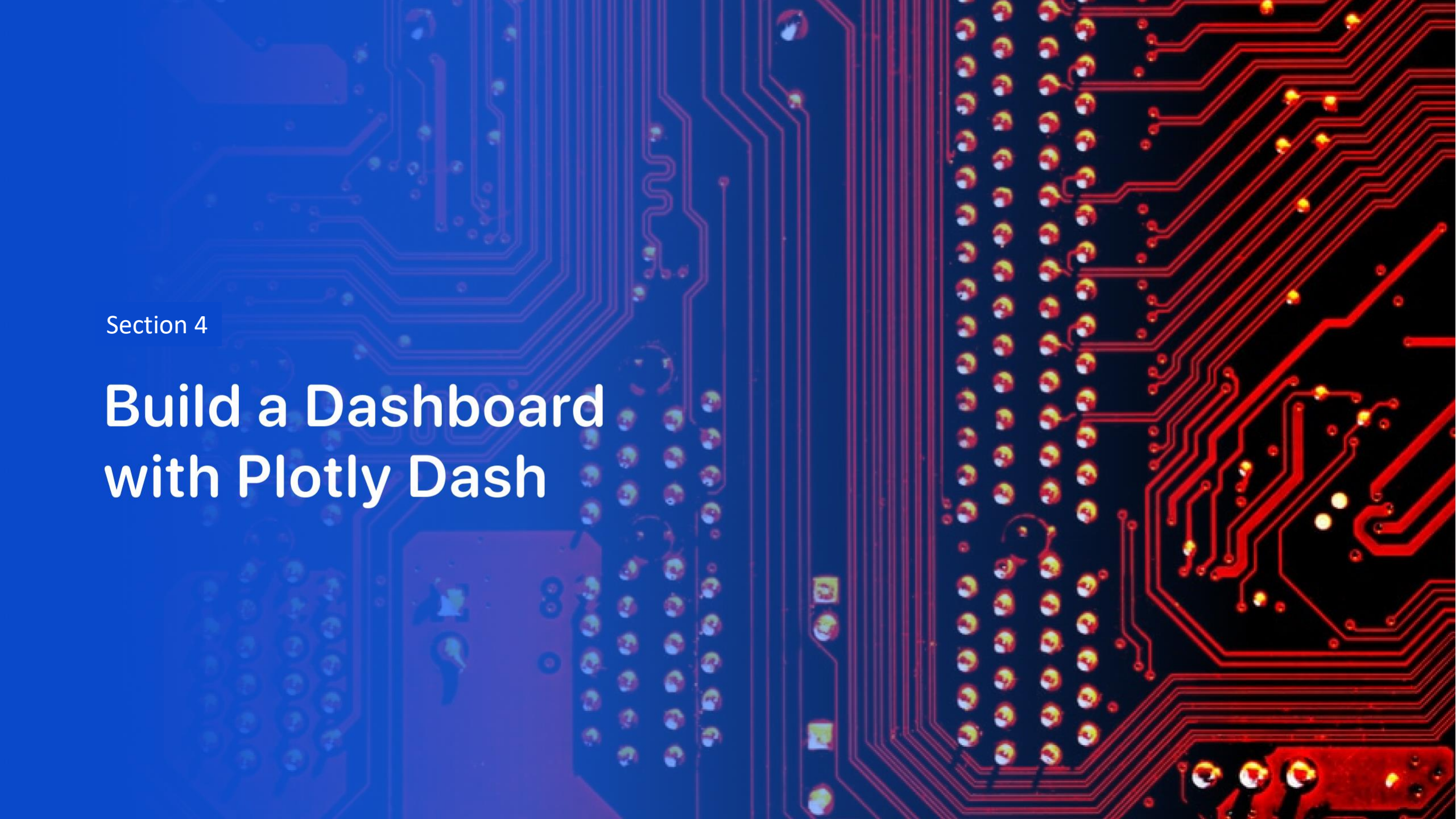
Folium Map – Color Labeled Launch Outcomes

- The folium map shows the color-labeled launch outcomes on the map
- You can see once you click on a specified location, it shows you the occurrences of successful and failed launches colored in green and red.

Folium Map – Proximity Lines

- The generated folium map shows a line from a selected launch site to its proximities such as railway, highway, coastline, or city with distance calculated and displayed



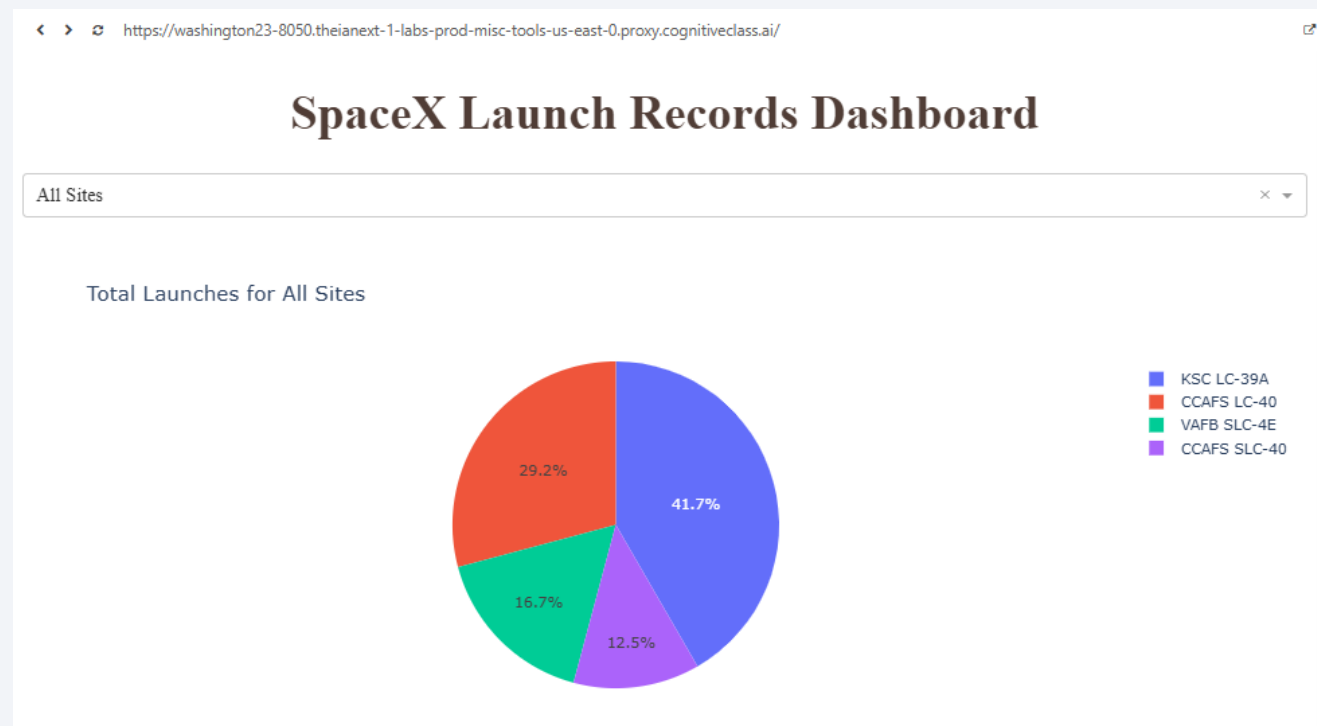


Section 4

Build a Dashboard with Plotly Dash

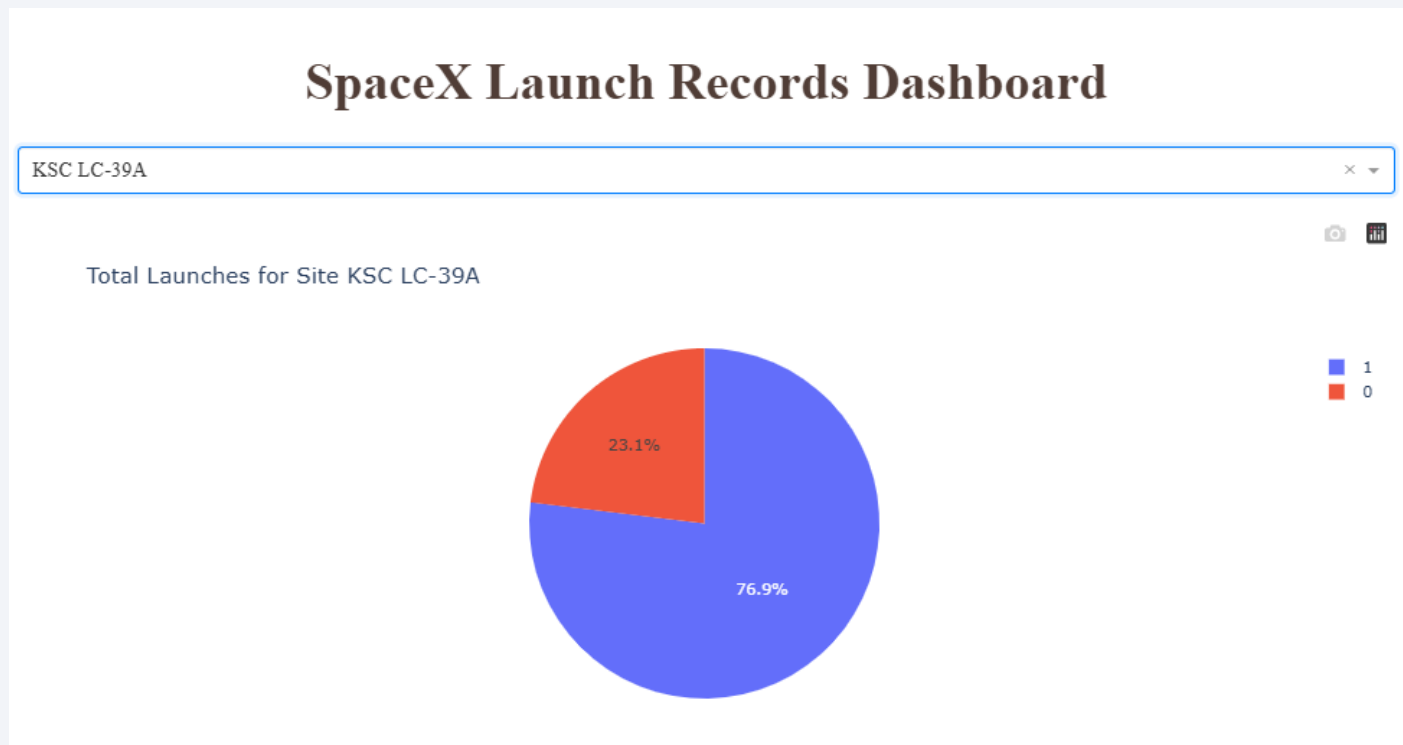
Launch Success Count by Site

- The dashboard shows launch success count for all sites, in a piechart
 - Of the 4 launch sites, the 'KSC LC-39A' had the highest number of successful launches

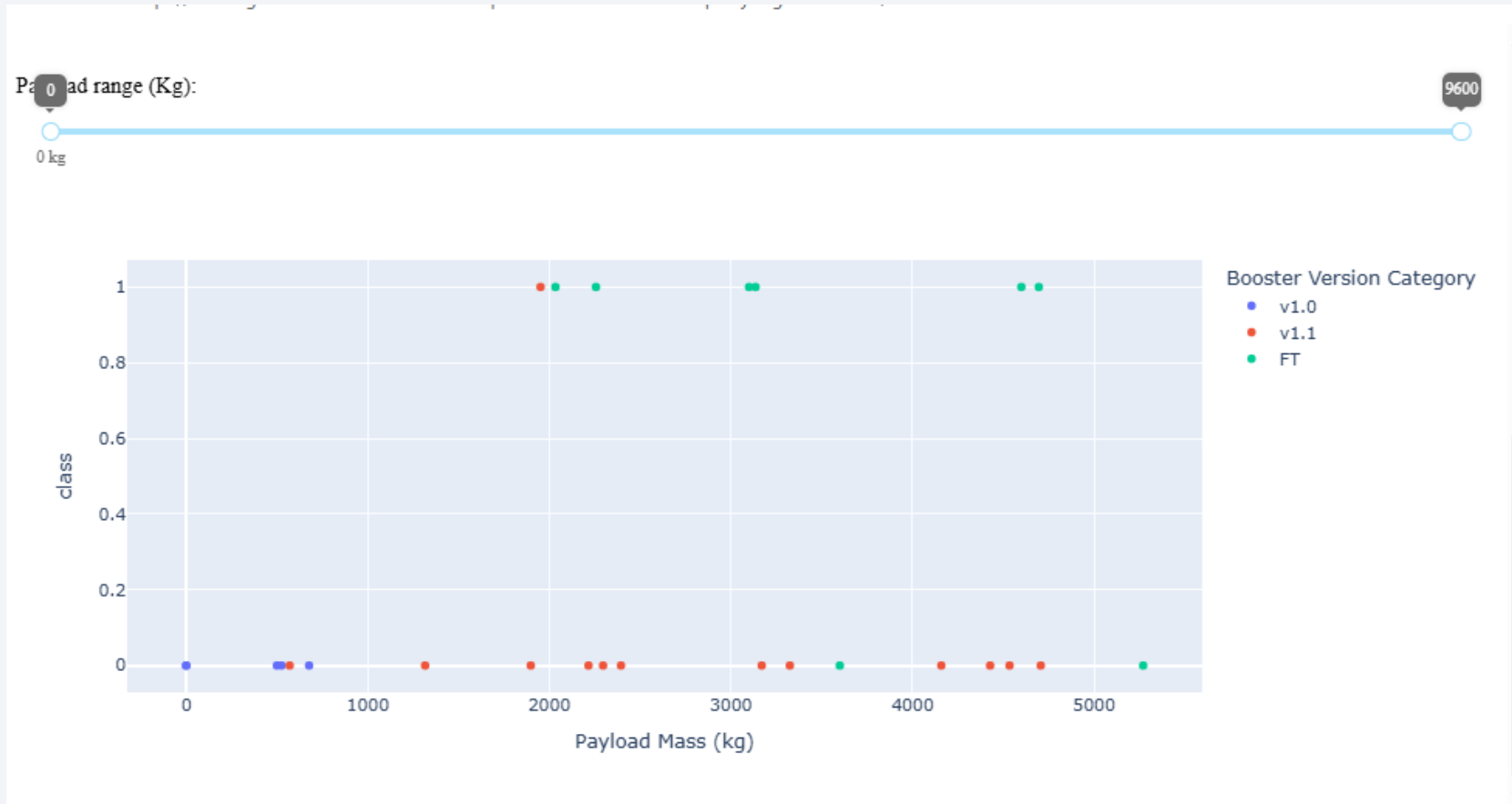


Success vs. Failed Launches – KSC LC-39A

- Pie chart shows the launch site with highest launch success ratio
 - The success rating at the KSC LC-39A launch site is roughly 77%



Payload vs. Launch Outcome



- Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider

- Here, you can see the relation of Payload Mass and, Booster Version, and Launch Outcome. We see that higher payload ranges and booster version FT have higher chances of success.

Section 5

Predictive Analysis (Classification)

Classification Accuracy

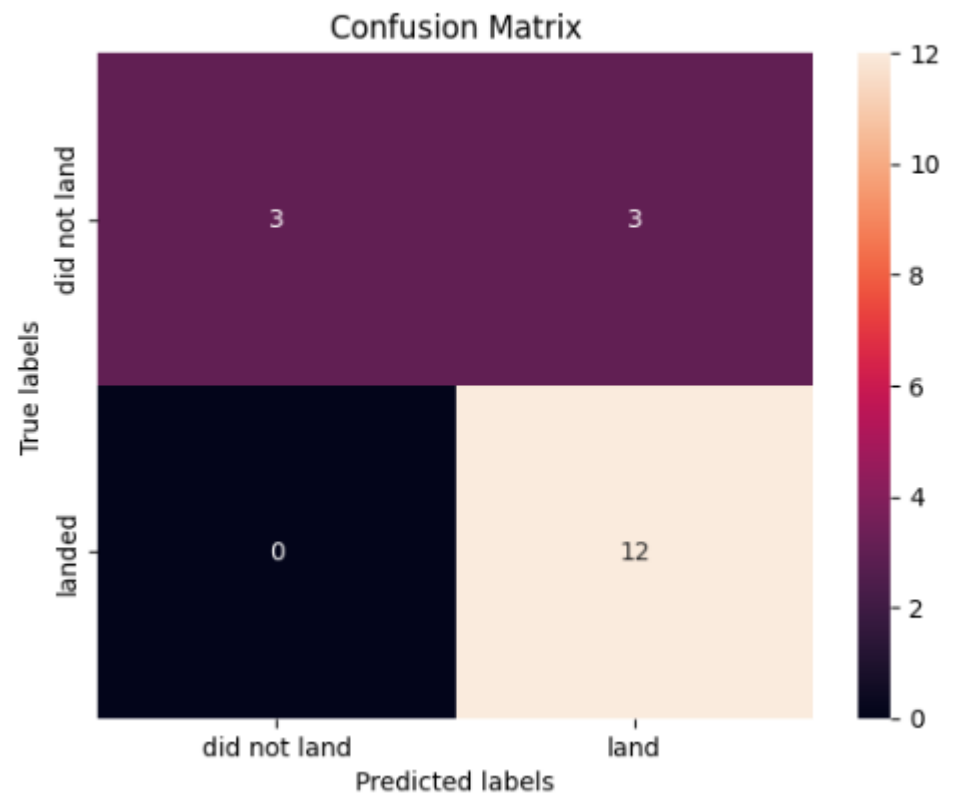
- Using Grid Search, I found that the tree model, using 'gini' as criterion, '2' as max depth, 'sqrt' as max_features, '2' as min_samples_leaf, '5' as min_samples_split, and 'best' as splitter, gave us the highest accuracy of .875.

```
print("tuned hyperparameters :(best parameters) ",tree_cv.best_params_)
print("accuracy :",tree_cv.best_score_)

tuned hyperparameters :(best parameters) {'criterion': 'gini', 'max_depth': 2, 'max_features': 'sqrt', 'min_samples_leaf': 2, 'min_samples_split':
5, 'splitter': 'best'}
accuracy : 0.875
```

Confusion Matrix

```
yhat = tree_cv.predict(X_test)
plot_confusion_matrix(Y_test,yhat)
```



- Examining the confusion matrix, we see that logistic regression can distinguish between the different classes. We see that the problem is false positives.
- True Positive - 12 (True label is landed, Predicted label is also landed)
- False Positive - 3 (True label is not landed, Predicted label is landed)

Conclusions

- Demonstrated application of data science methods to space sector
- Extracted meaningful insights from public SpaceX data
- Built machine learning models to predict Falcon 9 landing outcomes
- Random Forest classifier achieved highest accuracy of 84% Identified payload mass, booster version and site as important factors
- Models provide quantitative approach to pricing negotiations
- Insights on predictor effects can aid engineering of landings
- Public data and open source tools enabled robust problem solving
- Accuracy validation on test set showed models generalized well
- Overall, project achieved the goal of predicting SpaceX reusability

Appendix

[Jay-the-scientist/space-race \(github.com\)](https://github.com/Jay-the-scientist/space-race)

Thank you!

